

Aaron Dutta

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EDUCATION

University of Waterloo

BMath, Honours Mathematics with Co-Op

Waterloo, ON

Expected June 2031

EXPERIENCE

Mathematics Mentor

2024 – Present

Felix School of Education

- Mentoring several Grade 7-12 students through weekly math circles that encourage collaborative problem solving
- Advancing students' critical thinking skills by combining theoretical concepts with practical exercises

Swim Coach

2023 – Present

Olympian Swimming

- Delivering structured swim lessons focused on skill development, swim posture, and stroke technique
- Maintaining attendance and evaluating progress through comprehensive report cards to track growth

PROJECTS

ClipCoach | *Python, FastAPI, Next.js, React, TypeScript, Tailwind, librosa, ffmpeg, Pydantic*

[GitHub](#)

- Built a **multimodal highlight engine** that fuses **librosa audio-energy** and **ffmpeg motion** signals into a weighted excitement curve with **greedy peak selection** and minimum clip spacing
- Shipped a **FastAPI** pipeline with staged job orchestration, **ffmpeg cut/concat + music mixing**, and a fast **re-render path** that skips re-analysis when users edit clip timelines
- Delivered a **Next.js/React** upload-to-preview flow with progress tracking, embedded player, and **timeline editor** (keep/drop/reorder/nudge); deployed frontend on **Vercel** and API on **Railway**

FraudGen | *Python, Anthropic API, FastAPI, Streamlit, Pandas, NetworkX, Pydantic*

[GitHub](#)

- Designed a **4-agent adversarial LLM pipeline** that explores fraud variant space via a **coverage matrix** and generates **persona-seeded mule-network** scenarios
- Shipped **async parallel orchestration**, **deterministic validation/repair**, and **critic scoring** with retry loops and **ML-ready exports** for **XGBoost/GNN** training
- Built a **Streamlit** monitoring console with **network graph visualization** and dataset browser; exposed the pipeline through **FastAPI REST endpoints**

TechniqueTitan | *Python, OpenCV, MediaPipe, NumPy, Streamlit, PyYAML, pytest*

[GitHub](#)

- Built an end-to-end **computer vision pipeline** that extracts **21 hand landmarks** per hand with **MediaPipe**, normalizes coordinates for scale invariance, and scores **5 piano posture criteria** via geometry-based heuristics
- Shipped a **Streamlit UI** with photo, video timeline, and live camera modes plus a **batch CLI** exporting **CSV/JSON** metrics; scoring thresholds are **YAML-configurable** without code changes
- Implemented bilateral **hand disambiguation**, severity-coded skeleton overlays, and a **pytest** suite across geometry, feature extraction, and analysis modules

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML5/CSS3, SQL

Frameworks/Platforms: FastAPI, Flask, Next.js, React, Streamlit, Tailwind CSS

Technologies/Tools: Anthropic API, Docker, ffmpeg, Figma, Gemini API, Git, Google Maps API, librosa, MediaPipe, NetworkX, NumPy, OpenCV, Pandas, Pydantic, PyYAML, Railway, RESTful APIs, SciPy, SQLite, Supabase, Vercel

CERTIFICATIONS

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| • Google AI Essentials Specialization | 2026 |
| • Anthropic's Building with the Claude API | 2026 |
| • Harvard University's CS50x | 2024 |